3/8/2026
Ex-4:

SETTING UP A LOCAL AREA NETWORK (LAN)

AIM:-

To design, setup and configure a local area network (LAN) connecting four computers through a network switch and demonstrate resource / data sharing across the network

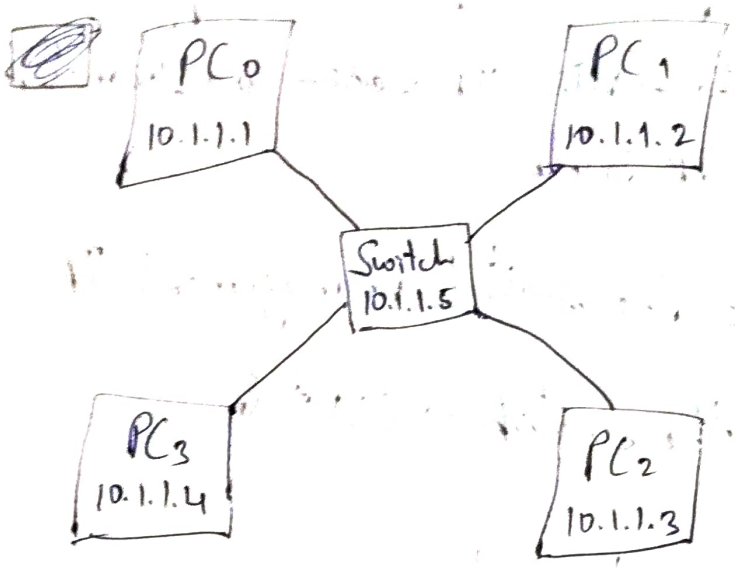
EQUIPMENT REQUIRED:-

Equipment	Qty	Specification
Personal computer	4	Ethernet(NIC) enabled
Network switch	1	8/16/24 port
Ethernet cables (cat5e/cat6)	4	straight-through, RJ45 connectors.

IP Address configuration table :-

Device	IP address	subnet mask	connected to
PC0	10.1.1.1	255.0.0.0	switch port fa0/1
PC1	10.1.1.2	255.0.0.0	switch port fa0/2
PC2	10.1.1.3	255.0.0.0	switch port fa0/3
PC3	10.1.1.4	255.0.0.0	switch port fa0/4
Switch	10.1.1.5	255.0.0.0	VLAN.1 interface

NETWORK DIAGRAM:



Implementation in CISCO packet tracer

1) Place the Devices

- Open CISCO packet Tracer
- From the device category panel, select End devices and drag 4PCs onto the workspace
- Select switches and drag one switch.

2) Cable the devices

- Click connections → select copper straight-through cable
- Click each PC, then click the switch, and select an available Fast ethernet port to completet each connection
- Repeat for all four PCs (ports Fa0/1 to Fa0/4). Green link lights confirm a physical successful condition.

3) Assign IP Addresses to PCs

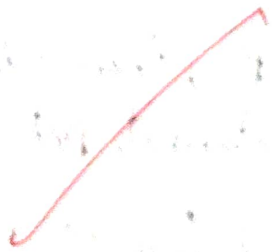
- Click apc → Desktop tab → IP configuration
- Enter the Static IP address and subnet mask as per the IP table.

4) Configure the switch management IP

- Click the switch → CLI tab
- enable
  configure terminal
  interface vlan 1
  ip address 10.1.1.5 255.0.0.0
  no shutdown
  exit

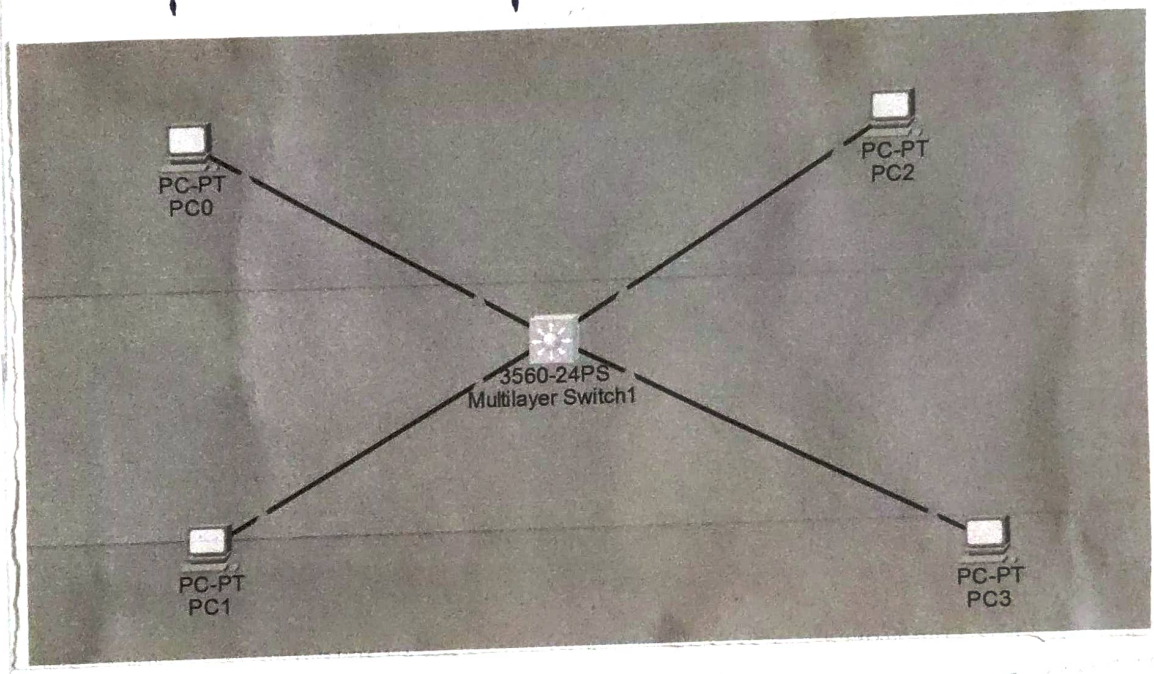
5) Verify connectivity using ping

- Click a PC → desktop tab → cmd prompt
- Use the ping to test connectivity
- Successful message.



6) Demonstrate Data Sharing (server simulation)

- Add a server device (End devices → Server-PT) and connect it to the switch.
- Assign it an IP address
- Open the server's services tab & enable the HTTP
- From another PC, open web browser and enter the server's IP address to load the page confirming successful data transfer across the LAN.





RESULT:

Thus LAN is settled up and connection is verified successfully.